



جامعة عين شمس
كلية الحاسبات والمعلومات

Project Allocation

Project Title :

**Vision-Based Obstacle Detection and Avoidance
System for an Autonomous Car Prototype**

Department: Scientific Computing

Supervisors:

1	Name	Department	Signature
1	Dr. Manal Tantawi	Scientific Computing	
2	TA. Aya Nasser	Scientific Computing	

Students Names:

No	Name	Department	Email
1	اسراء بسيوني محمد شرف	Scientific Computing	2022170054@cis.asu.edu.eg
2	سارة اسلام محمد محمود	Scientific Computing	2022170178@cis.asu.edu.eg
3	زياد خالد إبراهيم عبد الشافي	Scientific Computing	2022170164@cis.asu.edu.eg
4	عمرو محمد إبراهيم محمد	Scientific Computing	2022170291@cis.asu.edu.eg
5	زياد ياسر موسى موسى	Scientific Computing	2022170174@cis.asu.edu.eg

Introduction:

In recent years, autonomous driving has emerged as one of the most transformative fields in intelligent transportation systems. The increasing number of road accidents caused by human error, including distractions, speeding, and fatigue, has highlighted the urgent need for safer and more reliable driving technologies. Self-driving cars offer a potential solution by leveraging artificial intelligence, computer vision, and machine learning models to perceive the environment and make real-time decisions.

Our project focuses on building a self-driving car system capable of detecting road obstacles and traffic signs, and responding appropriately to ensure safe navigation. Unlike traditional approaches that rely mainly on sensors, our system emphasizes the use of camera-based vision models, which can not only detect obstacles but also classify their type (e.g., pedestrian, vehicle, or traffic sign). This enables the car to take intelligent actions, such as stopping, steering to the left or right, or reducing speed depending on the situation. Additionally, the system integrates real-time traffic sign recognition using trained object detection models, enabling compliance with road rules. By combining vision-based perception with intelligent decision-making, the project aims to replicate essential human driving behaviors in an automated and consistent manner, reducing the risk of accidents and improving road safety.

- Problem Definition:

Conventional vehicles rely entirely on human drivers to perceive the environment and react to sudden obstacles or traffic signs. This dependency on human attention and reaction time contributes significantly to road accidents worldwide. Although modern vehicles have introduced advanced safety features, accidents remain a major concern, with many collisions still occurring due to limitations in perception and delayed decision-making. Existing autonomous solutions are often complex and costly, restricting their widespread adoption and making them inaccessible in many contexts. The main challenge lies in enabling a vehicle to perceive its surroundings accurately and react safely in real time. Obstacles such as pedestrians or other vehicles may appear unexpectedly, requiring immediate responses like stopping or steering. Similarly, failure to recognize and obey traffic signs can lead to severe consequences. Developing a system that integrates obstacle detection, avoidance maneuvers, speed control, and road sign recognition into a unified vision-based framework is a critical problem to address.

- Motivation:

- Road accidents caused by human error account for the majority of traffic fatalities, creating a strong motivation to develop safer autonomous systems.
- An intelligent vehicle that can detect and avoid obstacles dynamically has the potential to significantly reduce collision risks and enhance passenger and pedestrian safety

- Real-time road sign detection ensures compliance with traffic regulations, improving overall driving reliability.
- Combining vision-based perception with sensor data provides a cost-effective yet powerful solution that can be adapted to both research and real-world deployment.
- By adopting a camera-first approach with trained computer vision models, our system can recognize both the existence and the identity of objects, allowing it to adjust behavior intelligently and contribute to safer, more reliable autonomous transportation.

Objectives :

The goal of this project is to develop a self-driving car system in simulation that can detect road obstacles and traffic signs using artificial intelligence and make safe driving decisions automatically. The system is also designed to assist the driver by taking correct actions in situations where the driver is not fully attentive.

The key objectives are:

- Detect and recognize obstacles, pedestrians, and traffic signs from camera input in real time.
- Enable the system to take appropriate actions such as stopping, slowing down, or steering to avoid collisions.
- Provide driver-assistance capability by acting as a backup when the driver is distracted or not fully focused.
- Evaluate the system in a simulation environment with different traffic, weather, and road conditions.
- Ensure safe navigation by reducing risks of accidents and improving compliance with road rules.

Time Plan:

Task	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Literature Review								
Data Preprocessing								
				FINAL EXAMS				
Obstacle Detection model								
Decision Making model								
Model Training and Evaluation								
Software & Hardware integration								
Documentation								

Tools :

Programming Languages: Python (TensorFlow, PyTorch, OpenCV), C++ (ROS integration)

Libraries: NumPy, Matplotlib, Scikit-learn, Pandas

AI Frameworks: TensorFlow, PyTorch, Keras

Simulation Platforms: CARLA Simulator, SVL Simulator
, ROS (Robot Operating System)

Computer Vision: OpenCV

Data Visualization: Matplotlib, Seaborn

Development Environment: Jupyter Notebooks, PyCharm, or VSCode

Version Control: Git/GitHub

Hardware components: Raspberry Pi 4, Arduino/ESP32, Camera Module, Ultrasonic Sensors, IR Sensors, DC Motors with L298N Driver

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Approvals :

Supervisor

(Dr. Manal Tantawi)

Head of Department

(Prof. Dr. Howida Shedeed)

Vice Dean of Education

(Prof. Dr. Hala Mosheir)

Dean of FCIS

(Prof. Dr. Rasha Ismail)